

Exercise 1 Task 2: ER Model Description

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Problem A: Personal Task Management

The model describes a personal task management system based on a digital sticker board. The board contains urgency-based areas. Each sticker represents exactly one task. Tasks may have responsible persons and may be connected to other tasks by a sequence or dependency relation.

Entity	Role in the model
StickerBoard	Container for areas and tasks displayed as stickers.
Area	Urgency-based group on the board.
Sticker	Visual object that represents exactly one task.
Task	Main work item with title, description, status, and deadline.
Status	Controlled set of task states such as planned, in progress, or completed.
Person	Responsible person who can be assigned to tasks.
TaskAssignment	Associative entity for optional many-to-many task-person assignment.
TaskDependency	Associative entity for predecessor/successor task relationships.

Major relationships

- A StickerBoard contains one or more Areas.
- An Area may contain zero or more Stickers.
- Each Sticker represents exactly one Task, and each Task has exactly one Sticker in this model.
- Each Task has one Status. A Status can be used by many Tasks.
- A Task may have zero or more responsible Persons, represented through TaskAssignment.
- A Task may precede or follow other Tasks, represented through TaskDependency.

Relation to existing applications

The model is inspired by common task management applications such as Trello, Todoist, Notion, and Asana. Trello is especially close to the board/list/card metaphor. Todoist and Asana show useful examples of deadlines, responsible people, status values, and task dependencies.

Limitations

The ER model focuses on persistent data. It does not describe screen operations such as drag and drop, notifications, reminders, or detailed access control. These aspects would require additional system design models.